Forecasting with Machine Learning: Introduction

Forecasting with Machine Learning

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2026-10-26

Introduction

Outline for Today

Course information Content, roadmap, format, and how to read the papers

Forecasting problems and data structures Targets, horizons, time-series, cross-sectional, and panel data

Forecasting information and predictors Lags, firm characteristics, mixed frequencies, feature construction

Forecasting objectives and machine learning Loss functions, optimal forecasts, flexibility and overfitting

Why Does ML Work? Variable Selection and Interpretation

Course Information

Forecasting with Machine Learning

This course combines modern machine learning methods with economic and financial forecasting

COMPONENT 1

Forecasting foundations

Construction, evaluation, and comparison of forecasts

COMPONENT 2

Machine learning methods

Shrinkage, tree-based methods, neural networks, and Python implementation

COMPONENT 3

Empirical applications

Research articles on stock returns, macroeconomics, volatility, and energy markets

> The emphasis is on understanding how ML methods are used in actual forecasting problems, rather than on learning algorithms in isolation

Course Roadmap

Week	Main topics	Lecturer
1	Forecasting foundations: construction, evaluation, and comparison	BV
2	Python programming and neural networks	NV
3	Machine learning for stock return prediction	NV
4	Machine learning for macroeconomic forecasting and inflation	BV
5	High-frequency econometrics and volatility forecasting	KC
6	Machine learning for volatility forecasting; mixed-frequency nowcasting	BV
7	Electricity and crude oil forecasting	KC

5 ECTS, seven weeks, two lectures of two hours per week

Teaching Format and Materials

Teaching format

- Traditional lectures and discussions
- Python demonstrations and selected exercises
- Applications based on research articles

Literature, software, and materials

- Selected research articles are the main teaching material; references are provided for each topic
- Python is the primary programming language
- Slides, code, and exercises are available on the course website and university platform

Learning from Research Articles

The course uses research papers to study how ML is applied to real forecasting problems. When reading a paper, consider:

1	**Forecasting problem**	What is predicted, and at what horizon?
2	**Data and predictors**	What information is used?
3	**Methods**	Which ML methods are considered, and why?
4	**Evaluation**	How are forecasts evaluated, and against which benchmarks?
5	**Findings**	What do we learn about the usefulness of machine learning?

> Focus on the main ideas and empirical findings, as well as the technical details

About Myself

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- Research interests: financial econometrics (high-frequency data), machine learning, microeconometrics, and quantitative finance
- Teaching interests: econometrics, finance, machine learning, and mathematics
- Faculty at Aarhus University (ECON) since 2014; professor since 2025
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Forecasting Problems and Data Structures

Why Forecast?

Forecasting plays a central role in economic and financial decision-making. Examples from this course:

- **Stock returns** Which stocks are likely to perform better? Implications for portfolio allocation
- **Macroeconomics** How will inflation and economic activity evolve? Implications for economic policy
- **Volatility** How risky will financial markets be? Implications for risk management
- **Energy markets** What will electricity or oil prices be? Implications for trading and planning

> The forecasting objective differs considerably across these applications

The Forecasting Problem

At time t, we want to forecast an outcome Y_{t+h}, where h is the **forecast horizon**. We have an information set $\mathcal{I}_t$ containing information available at time t. A forecasting method produces

$$\widehat{Y}_{t+h|t} = f_t(\mathcal{I}_t)$$

TARGET

What do we forecast?

HORIZON

How far ahead?

INFORMATION

What is known when the forecast is made?

> A genuine forecast cannot use information that becomes available after the forecast origin

What Exactly Are We Forecasting?

Application	Typical target	Example
Stock returns	Future return	Next month’s stock return
Macroeconomics	Growth rate or change	Quarterly inflation or GDP growth
Volatility	Future variance or volatility	Next day’s realized variance
Energy	Price level or return	Tomorrow’s hourly electricity prices; future crude oil returns

- Forecasting a price level is different from forecasting its return or change
- Forecasting volatility is different from forecasting the direction of returns

> The definition of the target influences the choice of models, predictors, and evaluation

Time-Series, Cross-Sectional and Panel Forecasting

TIME-SERIES FORECASTING

Predict future observations of a variable using historical information

Example: US inflation over time

CROSS-SECTIONAL PREDICTION

Predict differences across individual units

Example: which stocks are expected to outperform others next month?

PANEL FORECASTING

Observations for many units over time: $Y_{i,t}$, $i = 1, \ldots, N$, $t = 1, \ldots, T$

Example: monthly returns for thousands of individual stocks

Important distinction: Panel forecasting can involve both time-series prediction and cross-sectional comparisons

Example: Predicting Individual Stock Returns

Suppose we observe monthly data for a large number of stocks. For each stock i at time t, the objective is to predict next month's return,

$$\hat{r}_{i,t+1|t} = f(X_{i,t})$$

where $X_{i,t}$ contains predictors available for stock i at time t.

Stock	Month	Predictors	Target
A	January	$X_{A,\text{Jan}}$	$r_{A,\text{Feb}}$
B	January	$X_{B,\text{Jan}}$	$r_{B,\text{Feb}}$
A	February	$X_{A,\text{Feb}}$	$r_{A,\text{Mar}}$
B	February	$X_{B,\text{Feb}}$	$r_{B,\text{Mar}}$

Example: Predicting Individual Stock Returns II

- The model uses information across both stocks and historical months
- Predictions can be evaluated for individual stocks or used to rank stocks for portfolio construction

Different Forecasting Problems, Different Challenges

Stock returns Returns contain substantial noise; predictive signals are often weak

Inflation Inflation can be persistent, but predictive relationships may change across economic conditions

Volatility Volatility exhibits strong persistence and clustering over time

Electricity prices Strong daily and weekly patterns, together with occasional extreme price movements

Different forecasting problems have different data structures and sources of predictability.
No single forecasting method is necessarily best across all applications

Forecasting Information and Predictors

Where Does Predictive Information Come From?

A forecasting model uses observable information to predict an unknown future outcome,

$$\widehat{Y}_{t+h|t} = f(X_t)$$

where X_t denotes the available predictors.

Where Does Predictive Information Come From? II

Predictive information may come from:

Past values of the target

Lagged inflation, volatility, or electricity prices

Other economic variables

Interest rates, employment, economic activity, and firm characteristics

Constructed features

Moving averages, momentum measures, technical indicators, and interactions

Alternative data

Internet searches, news, or financial text

The usefulness of each information source depends on the forecasting problem

Persistence and Lagged Predictors

Many economic and financial variables exhibit persistence: past values contain information about future outcomes. A simple autoregressive forecasting model is

$$Y_{t+1} = \alpha + \sum_{j=0}^{p-1} \beta_j Y_{t-j} + \varepsilon_{t+1}$$

Machine learning can use the same lagged inputs in a more flexible forecasting function,

$$\widehat{Y}_{t+1|t} = f(Y_t, Y_{t-1}, \ldots, Y_{t-p+1}, Z_t)$$

Persistence and Lagged Predictors II

Examples:

- **Inflation** Past inflation and other macroeconomic indicators
- **Volatility** Recent realized volatility
- **Electricity prices** Prices from previous hours and days

> The relevance of lags depends on the variable, the horizon, and the stability of the underlying relationships

Example: Volatility Persistence

- Financial market volatility exhibits substantial persistence: high-volatility periods often follow other high-volatility periods
- The HAR model uses realized volatility measured over different historical horizons
- Let RV_t denote daily realized variance. Construct weekly and monthly averages,

$$RV_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} RV_{t-j}, \qquad RV_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} RV_{t-j}$$

Example: The HAR Model

A simple HAR forecasting specification is

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$$

ML models can use these predictors while allowing nonlinear relationships and interactions

> **Application**: Christensen et al. (2023) compare ML with HAR-type models

Example: Firm Characteristics and Stock Returns

In stock return prediction, explanatory variables often differ across firms as well as over time. Common firm characteristics include:

- **Size** Market capitalization
- **Valuation** Book-to-market and related ratios
- **Momentum** Past stock returns over selected horizons
- **Risk/trading activity** Volatility, turnover, liquidity measures

Example: Firm Characteristics and Stock Returns II

The forecasting model combines these variables,

$$\hat{r}_{i,t+1|t} = f(\text{Size}_{i,t}, \text{Momentum}_{i,t}, \text{Volatility}_{i,t}, \ldots)$$

Predictors can also interact with aggregate economic and market conditions

> **Application**: Gu et al. (2020) examine a large collection of firm characteristics and macroeconomic interactions

Predictors Observed at Different Frequencies

Tasks often involve information observed at different frequencies. Examples:

- Daily financial market data
- Weekly unemployment claims
- Monthly inflation and employment indicators
- Quarterly GDP

Mixed-frequency forecasting seeks to exploit information without necessarily reducing all predictors to the lowest frequency

Mixed Frequencies and Nowcasting

- **Nowcasting**: estimating a current or very recent economic outcome before its official value becomes available
- Example: Borup et al. (2023) use daily internet search data to nowcast and backcast weekly US initial unemployment claims

> **Important**: The date an observation refers to and the date it becomes available are not necessarily the same

Feature Construction: Turning Data into Predictors

Raw observations can be transformed into predictors that summarize relevant patterns. Common feature constructions:

- **Lags**: $Y_t, Y_{t-1}, Y_{t-2}, \ldots$
- **Moving averages**:

$$Y_t^{(k)} = \frac{1}{k} \sum_{j=0}^{k-1} Y_{t-j}$$

- **Changes and returns**: differences or percentage changes instead of levels
- **Calendar variables**: hour of day, day of week, or season

Feature Construction II

Examples from energy forecasting:

Electricity

- Lagged prices
- Demand forecasts
- Calendar effects

Crude oil

- Macroeconomic variables
- Technical indicators

> Features must be constructed using information available at the forecast origin

Forecasting Objectives and Machine Learning

Forecast Errors and Loss Functions

A forecasting method produces a prediction $\widehat{Y}_{t+h|t}$ of the future outcome Y_{t+h}. Once the outcome is observed, the forecast error is

$$e_{t+h|t} = Y_{t+h} - \widehat{Y}_{t+h|t}$$

Forecasting methods aim to minimize some measure of forecast error, represented by a **loss function**,

$$L(Y_{t+h}, \widehat{Y}_{t+h|t}) = L(e_{t+h|t})$$

> Different loss functions penalize forecasting mistakes differently.
> The choice of loss function determines what the forecasting model seeks to predict

Squared and Absolute Error Loss

Two commonly used loss functions are:

Squared error loss

$$L(e) = e^2$$

Large errors receive disproportionately greater penalties

Absolute error loss

$$L(e) = |e|$$

Loss increases linearly with the magnitude of the error

For example, increasing the absolute error from 1 to 3:

- Squared loss increases from 1 to 9
- Absolute loss increases from 1 to 3

> Squared loss is more sensitive to extreme forecast errors than absolute loss

Optimal Forecasts under Squared Loss

Suppose we want to forecast Y_{t+h} given information $\mathcal{I}_t$. Under squared loss, the optimal forecast f_t^* solves

$$\min_f \mathrm{E}\left[(Y_{t+h} - f)^2 \mid \mathcal{I}_t\right]$$

The first-order condition is

$$\mathrm{E}\left[Y_{t+h} - f_t^* \mid \mathcal{I}_t\right] = 0$$

Optimal Forecasts under Squared Loss II

Hence,

$$f_t^* = \mathrm{E}\left[Y_{t+h} \mid \mathcal{I}_t\right]$$

- Under squared loss, the optimal point forecast is the conditional mean

> **Implication**: Regression-based ML methods trained using squared loss aim to approximate the conditional mean

Different Loss Functions, Different Forecasts

The optimal forecast depends on the loss function, not only on the available data.

Loss function	Optimal point forecast
Squared error	Conditional mean
Absolute error	Conditional median
Asymmetric absolute error	Conditional quantile

- Example: forecasting electricity prices in the presence of occasional extreme spikes
- Conditional means and medians may differ substantially when the predictive distribution is asymmetric or contains extreme outcomes
- Most standard regression-based ML applications use squared loss, but alternative objectives can be used when appropriate

From Linear Forecasting to Machine Learning

A traditional linear forecasting model:

$$Y_{t+1} = \alpha + \beta' X_t + \varepsilon_{t+1}$$

The forecast is a linear function of the predictors:

$$\widehat{Y}_{t+1|t} = \widehat{\alpha} + \widehat{\beta}' X_t$$

From Linear Forecasting to Machine Learning II

Machine learning allows more flexible forecasting functions,

$$\widehat{Y}_{t+1|t} = \widehat{f}(X_t)$$

Ridge and LASSO

Regularized linear models

Random forests

Nonlinearities and interactions

Neural networks

Flexible nonlinear functions

These methods can share the same forecasting objective while differing in how they approximate the predictive relationship

Model Flexibility and Overfitting

Flexible models can capture complex predictive relationships, but may also fit noise in the data. Two familiar problems:

Underfitting

- A model is too restrictive to capture relevant predictive patterns

Overfitting

- A model fits the training observations well but generalizes poorly to new observations

The bias–variance trade-off illustrates the tension between model flexibility and estimation uncertainty

Model Flexibility and Overfitting II

ML methods control complexity through:

- Regularization in linear models
- Restrictions on tree complexity
- Regularization and early stopping in neural networks

> **Central question**: when does additional model flexibility capture genuine predictive information rather than noise?

Why Does ML Work? Variable Selection and Interpretation

Where Do ML Forecasting Gains Come From?

Machine learning may improve forecasting through several distinct mechanisms:

1	**More predictive information**	Incorporating additional variables and constructed features
2	**Regularization and variable selection**	Controlling estimation uncertainty when many predictors are available
3	**Nonlinearities and interactions**	Capturing predictive relationships that linear models may miss
4	**Model selection and tuning**	Choosing specifications and hyperparameters based on predictive performance

These mechanisms are related, but conceptually different

> **Research connection**: Goulet Coulombe et al. (2022) investigate why ML improves macroeconomic forecasting, emphasizing nonlinearities, regularization, cross-validation, and loss functions

Many Predictors, But Which Ones Matter?

Modern applications often contain large collections of predictors. Examples:

- Stock returns: firm characteristics and macroeconomic variables
- Inflation: a large set of macroeconomic and financial indicators
- Oil returns: macroeconomic variables and technical indicators

Many Predictors, But Which Ones Matter? II

Including more predictors does not necessarily improve forecasting:

- Some predictors contain little information
- Many predictors are strongly correlated
- Estimation becomes less precise as model complexity increases

Many Predictors, But Which Ones Matter? III

A central question is whether we can identify a relatively small set of useful predictors, or exploit many predictors without overfitting

> **Applications**: Medeiros et al. (2021) study inflation forecasting with many predictors, while Zhang et al. (2019) examine LASSO-based predictor selection for crude oil returns

Variable Selection and Regularization

Consider a linear forecasting model with p predictors, $Y_{t+1} = \alpha + X_t'\beta + \varepsilon_{t+1}$. LASSO estimates coefficients by solving

$$\min_{\alpha,\beta} \left\{ \frac{1}{T} \sum_{t=1}^{T} (Y_{t+1} - \alpha - X_t'\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

The penalty shrinks coefficients and can set some exactly to zero. Two potential benefits:

- Reducing estimation uncertainty
- Identifying a smaller set of predictors

Variable Selection and Regularization II

- However, selection need not be stable: correlated predictors may substitute for one another

> **Important distinction**: Selecting a variable because it improves prediction does not establish that the variable has a causal effect on the outcome

Nonlinearities and Interactions

Linear forecasting models impose an additive relationship with constant coefficients,

$$\widehat{Y}_{t+1|t} = \alpha + \beta_1 X_{1,t} + \beta_2 X_{2,t}$$

Predictive relationships may instead involve nonlinearities and interactions,

$$\widehat{Y}_{t+1|t} = \alpha + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \gamma X_{1,t} X_{2,t}$$

The predictive effect of $X_{1,t}$ now depends on the value of $X_{2,t}$,

$$\frac{\partial \widehat{Y}_{t+1|t}}{\partial X_{1,t}} = \beta_1 + \gamma X_{2,t}$$

Nonlinearities and Interactions II

Examples:

- Relevance of financial predictor may depend on market conditions
- Macroeconomic links may differ between expansions and recessions

> Trees and neural networks can learn such relationships without specifying each interaction explicitly

How Can We Identify the Source of Forecasting Gains?

Suppose a flexible ML model outperforms a traditional forecasting regression. Is the improvement due to:

- Better predictors?
- A more flexible functional form?
- Regularization or model tuning?

How Can We Identify the Source of Forecasting Gains? II

Controlled comparisons can help distinguish these possibilities:

Comparison	Predictors	Main question
OLS vs. LASSO	Same	Does regularization help?
Linear vs. nonlinear ML	Same	Does flexibility improve prediction?
ML with vs. without new features	Different	Do new predictors add information?

> These help explain why one approach performs better, rather than merely documenting which model has the lowest forecast error

Variable Selection, Importance, and Interpretation

Three related but different questions:

Concept	Main question
Variable selection	Which predictors should enter the model?
Variable importance	Which predictors contribute most to the fitted model’s predictions or predictive performance?
Model interpretation	How do the model’s predictions change with particular predictor values?

Variable Selection, Importance, and Interpretation II

Examples of tools:

- LASSO for variable selection
- Permutation-based variable importance
- Partial dependence and accumulated local effects for interpretation

> Importance and interpretation depend on the fitted model and the data distribution
> Predictive importance does not imply economic/causal importance

What Do Forecasting Models Learn?

Application	Relevant questions	Research example
Stock returns	Which characteristics help predict returns? Are nonlinear interactions important?	Gu et al. (2020)
Inflation	Which macroeconomic indicators are informative? How nonlinear are the relationships?	Medeiros et al. (2021)
Volatility	How important are daily, weekly, and monthly volatility components?	Christensen et al. (2023)
Oil returns	Which variables are selected from a large predictor set?	Zhang et al. (2019)

What Do Forecasting Models Learn? II

- A forecasting model can be useful even when its predictions are difficult to interpret
- However, understanding its predictive structure can help assess robustness, improve models, and generate new economic hypotheses

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